

A business request for the café: Preparing an EC2 instance to host a website (challenge #1)

Task 1: Analyzing the existing EC2 instance

4, 5.

The screenshot displays the AWS Management Console for the EC2 service in the us-east-1 region. The main content area shows a table of instances with the following details:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
aws-cloud9-Ca...	i-043b2a94c2f3fbdef	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-44-200...

The left sidebar contains a navigation menu with the following sections:

- EC2
 - Dashboard
 - EC2 Global View
 - Events
 - Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
 - Images
 - AMIs
 - AMI Catalog
 - Elastic Block Store
 - Volumes

The bottom of the screen shows a Windows taskbar with the system clock at 1:30 PM on 25-Oct-2025.

Task 1.1: Answering questions about the instance

6-8.

AWS Academy Cloud Architecting - Module 4 Challenge Lab Questions

View questions in: [English](#)

Question 1: Is the instance in a public subnet?

☒ Yes
☐ No

Question 2: Does the EC2 instance have an IPv4 Public IP address assigned to it?

☒ Yes
☐ No

Question 3: What inbound TCP port numbers are open for this instance?

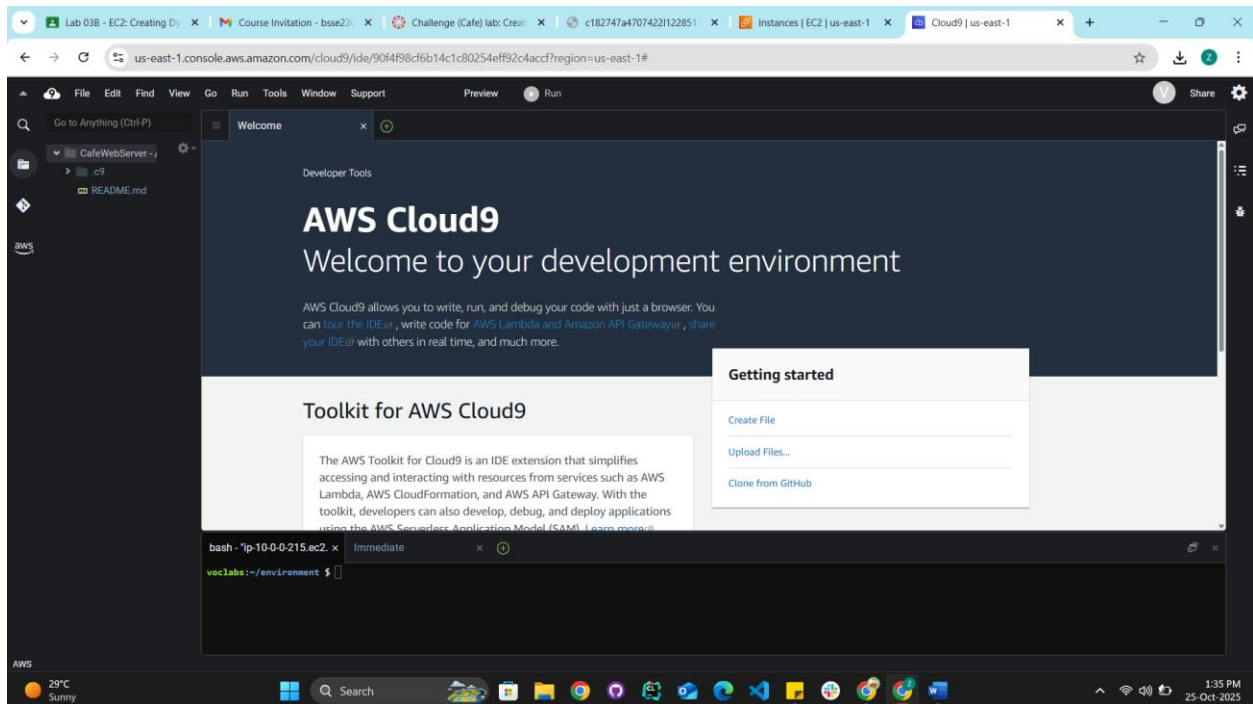
☐ TCP port 80 only, open to the internet
☐ TCP port 22 only, open to the internet
☐ TCP port 80 only, open to a specific range of IP addresses
☒ TCP port 22 only, open to a specific range of IP addresses

Question 4: Does the EC2 instance have an AWS Identity and Access Management (IAM) role associated with it?

☒ Yes
☐ No

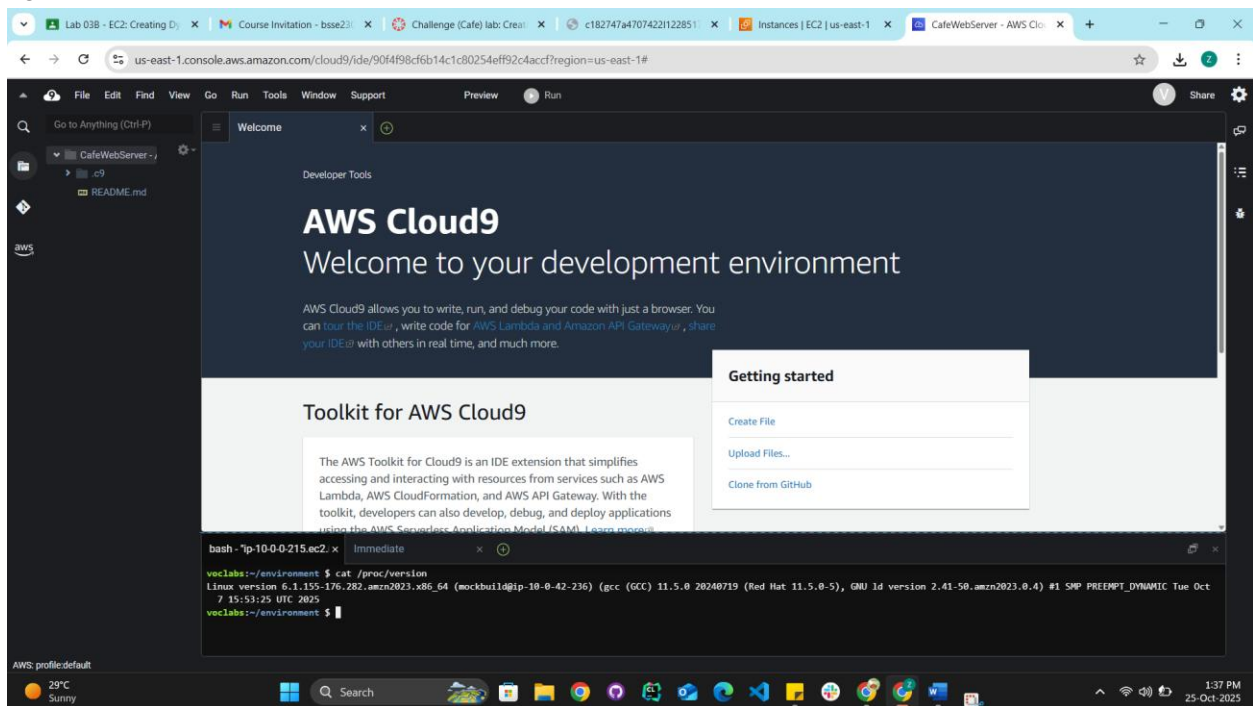
Task 2: Connecting to the IDE on the EC2 instance

9.

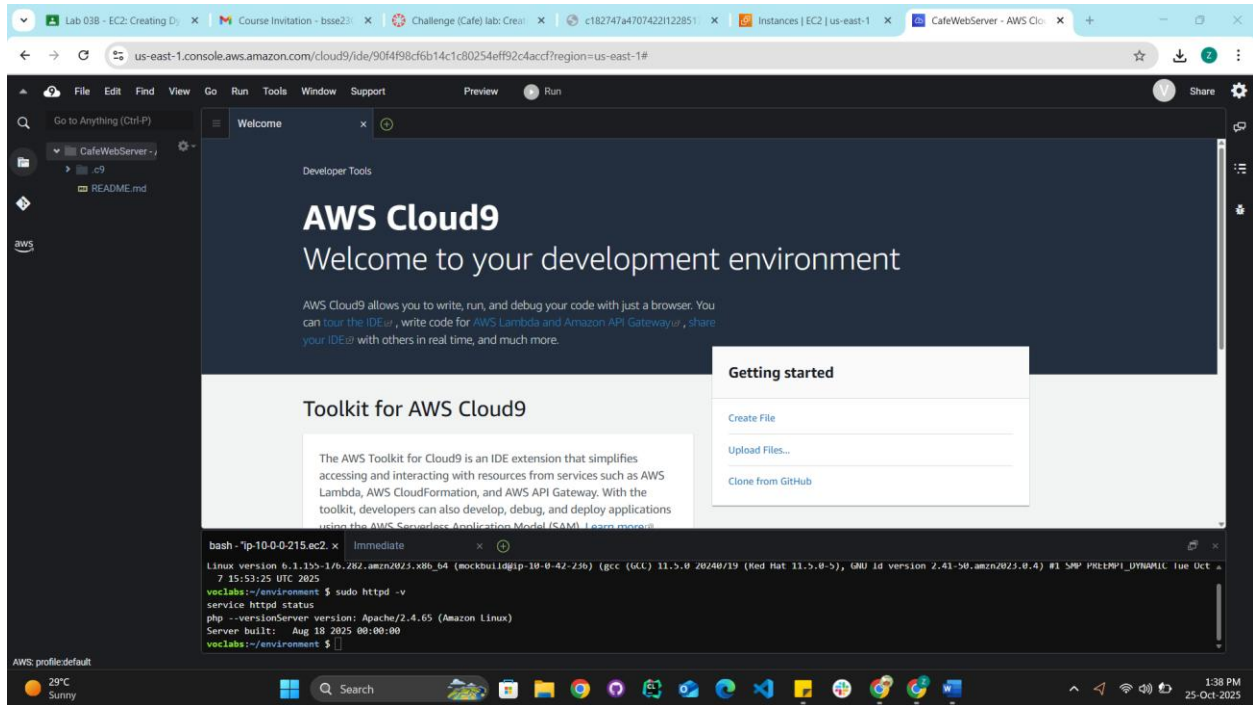


Task 3: Analyzing the LAMP stack environment and confirming that the web server is accessible

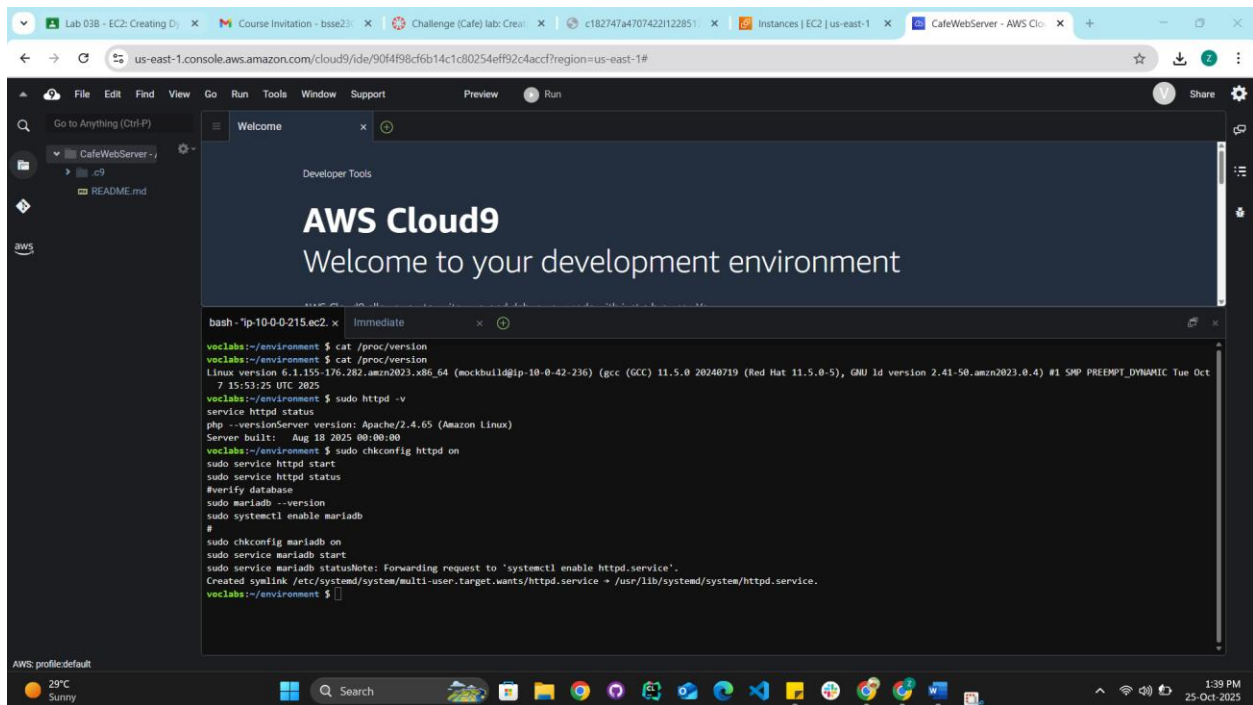
10.



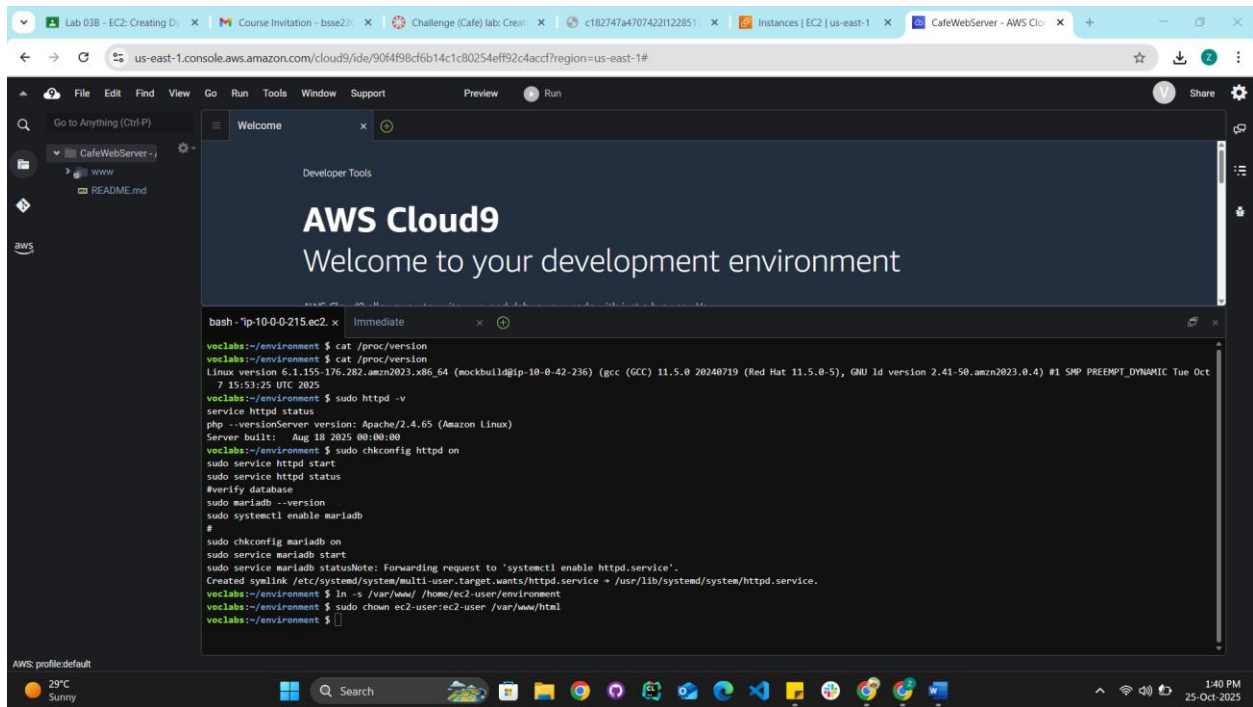
11.



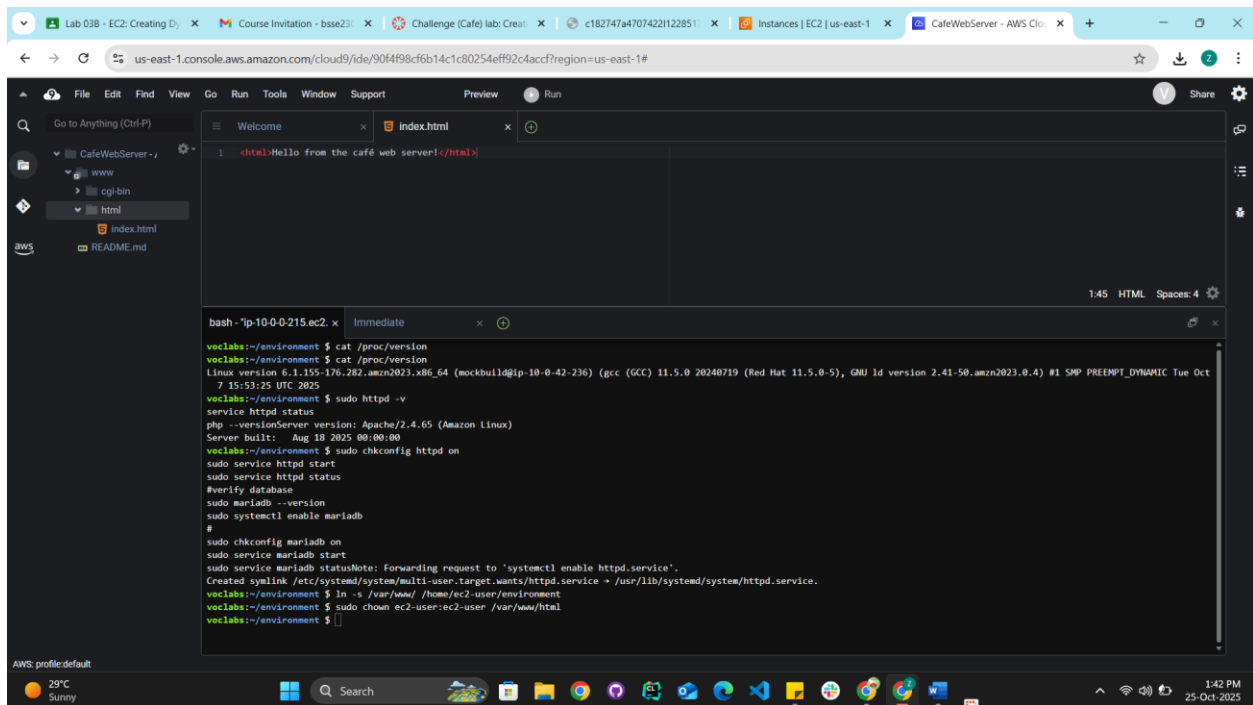
12.



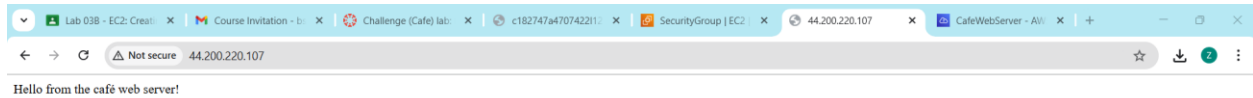
13.



14-17.



18.



New business requirement: Installing a dynamic website application on the EC2 instance (challenge #2)

Task 4: Installing the café application

19.

us-east-1.console.aws.amazon.com/cloud9/ide/90f4f98cf6b14c1c80254ef92c4accf?region=us-east-1#

```

1 <html>Hello from the cafe web server!</html>

```

```

bash -tp-10-0-0-215.ec2: x Immediate
inflat: /var/www/html/cafe/GuzzleHttp/Promise/PromiseInterface.php
inflat: /var/www/html/cafe/GuzzleHttp/Promise/TaskQueueInterface.php
inflat: /var/www/html/cafe/GuzzleHttp/Promise/CancellationException.php
inflat: /var/www/html/cafe/GuzzleHttp/Promise/RejectedPromise.php
inflat: /var/www/html/cafe/GuzzleHttp/Promise/Is.php
inflat: /var/www/html/cafe/GuzzleHttp/Promise/AggregateException.php
creating: /var/www/html/cafe/Psr/
creating: /var/www/html/cafe/Psr/Http/
creating: /var/www/html/cafe/Psr/Http/Message/
inflat: /var/www/html/cafe/Psr/Http/Message/ServerRequestInterface.php
inflat: /var/www/html/cafe/Psr/Http/Message/ResponseInterface.php
inflat: /var/www/html/cafe/Psr/Http/Message/MessageInterface.php
inflat: /var/www/html/cafe/Psr/Http/Message/UploadedFileInterface.php
inflat: /var/www/html/cafe/Psr/Http/Message/UriInterface.php
inflat: /var/www/html/cafe/Psr/Http/Message/StreamInterface.php
creating: /var/www/html/cafe/Psr/Http/Client/
inflat: /var/www/html/cafe/Psr/Http/Client/ClientInterface.php
inflat: /var/www/html/cafe/Psr/Http/Client/RequestExceptionInterface.php
inflat: /var/www/html/cafe/Psr/Http/Client/ResponseExceptionInterface.php
inflat: /var/www/html/cafe/Psr/Http/Client/NetworkExceptionInterface.php
inflat: /var/www/html/cafe/psr-autoloader.php
voclabs:/var/www/html/cafe $ chmod -R +r /var/www/html/cafe/
voclabs:/var/www/html/cafe $

```

20.

us-east-1.console.aws.amazon.com/cloud9/ide/90f4f98cf6b14c1c80254ef92c4accf?region=us-east-1#

```

1 <html>
2 <head>
3   <meta charset="UTF-8">
4   <title>Café</title>
5   <link rel="stylesheet" href="css/styles.css">
6 </head>
7
8 <body class="bodyStyle">
9
10  <div id="header" class="mainHeader">
11    <div class="center">Café</div>
12  </div>
13
14  <div class="topnav">
15    <a href="index.php">Home</a>
16    <a href="#aboutUs">About Us</a>
17    <a href="#contactUs">Contact Us</a>
18    <a href="#menu.php">Menu</a>
19    <a href="orderHistory.php">Order History</a>
20  </div>
21
22  <div id="mainContent">
23
24    <div id="mainPictures" class="center">
25      <table>
26

```

```

bash -tp-10-0-0-215.ec2: x Immediate
inflat: /var/www/html/cafe/Psr/Http/Client/ClientInterface.php
inflat: /var/www/html/cafe/Psr/Http/Client/RequestExceptionInterface.php
inflat: /var/www/html/cafe/Psr/Http/Client/NetworkExceptionInterface.php
inflat: /var/www/html/cafe/psr-autoloader.php
voclabs:/var/www/html/cafe $ chmod -R +r /var/www/html/cafe/
voclabs:/var/www/html/cafe $

```

21.

Lab 038 - EC2: Crea... Course Invitation - b... Challenge (Cafe) lab... c182747a470742211... SecurityGroup [EC2]... 44.200.220.107... CafeWebServer - All...
us-east-1.console.aws.amazon.com/cloud9/ide/90f4f98cf6b14c1c80254ef92c4accf?region=us-east-1#f

```
<?php
require 'aws.phar';
require 'aws-autoloader.php';
$ch = curl_init();

// get a valid TOKEN
$headers = array(
    'X-aws-ec2-metadata-token-ttl-seconds: 21600' );
$url = "http://169.254.169.254/latest/api/token";
curl_setopt($ch, CURLOPT_HTTPHEADER, $headers );
curl_setopt($ch, CURLOPT_RETURNTRANSFER, true );
curl_setopt($ch, CURLOPT_CUSTOMREQUEST, "PUT" );
curl_setopt($ch, CURLOPT_URL, $url );
$token = curl_exec( $ch );

echo "cp TOKEN : " . $token;
// then get metadata of the current instance
$headers = array (
    'X-aws-ec2-metadata-token: ' . $token );
$url = "http://169.254.169.254/latest/meta-data/placement/availability-zone";
curl_setopt($ch, CURLOPT_URL, $url );
curl_setopt($ch, CURLOPT_HTTPHEADER, $headers );
curl_setopt($ch, CURLOPT_RETURNTRANSFER, true );
curl_setopt($ch, CURLOPT_CUSTOMREQUEST, "GET" );
$result = curl_exec( $ch );
$az = curl_exec( $ch );

echo "cp RESULT : " . $result;
$region = substr($az, 0, -1);
```

bash -ip-10-0-0-215.ec2.x Immediate

```
inflating: /var/www/html/cafe/Psr/Http/Client/ClientInterface.php
inflating: /var/www/html/cafe/Psr/Http/Client/RequestExceptionInterface.php
inflating: /var/www/html/cafe/Psr/Http/Client/NetworkExceptionInterface.php
inflating: /var/www/html/cafe/aws-autoloader.php
voclabs:/var/www/html/cafe $ chmod -R sr /var/www/html/cafe/
voclabs:/var/www/html/cafe $
```

AWS profile: default
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22.

Lab 038 - EC2: Crea... Course Invitation - b... Challenge (Cafe) lab... c182747a470742211... Secrets | Secrets Man... 44.200.220.107... CafeWebServer - All...
us-east-1.console.aws.amazon.com/cloud9/ide/90f4f98cf6b14c1c80254ef92c4accf?region=us-east-1#f

```
<?php
require 'aws.phar';
require 'aws-autoloader.php';
$ch = curl_init();

// get a valid TOKEN
$headers = array(
```

bash -ip-10-0-0-215.ec2.x Immediate

```
{
  "ARN": "arn:aws:secretsmanager:us-east-1:315224400642:secret:/cafe/currency-CIIWV",
  "Name": "/cafe/currency",
  "VersionId": "1ca9397c-2f3f-4f89-b562-baf11dc7c21"
}
{
  "ARN": "arn:aws:secretsmanager:us-east-1:315224400642:secret:/cafe/dbUrl-CuoXJ1",
  "Name": "/cafe/dbUrl",
  "VersionId": "a9992aa5-5379-45a6-92cc-48c59b073530"
}
{
  "ARN": "arn:aws:secretsmanager:us-east-1:315224400642:secret:/cafe/dbName-cZnGX1",
  "Name": "/cafe/dbName",
  "VersionId": "16646cbe-5dee-4059-b95b-79d0e402e104"
}
{
  "ARN": "arn:aws:secretsmanager:us-east-1:315224400642:secret:/cafe/dbUser-FFXPTm",
  "Name": "/cafe/dbUser",
  "VersionId": "6f1e6165-c1e2-4e03-b6ba-e7286c5b2aa5"
}
{
  "ARN": "arn:aws:secretsmanager:us-east-1:315224400642:secret:/cafe/dbPassword-2uowG1",
  "Name": "/cafe/dbPassword",
  "VersionId": "8458b7e9-8b15-479b-b26b-9f8b33746434"
}
Application Secrets Setup script completed.
voclabs:/environment/setup $
```

AWS profile: default
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24.

us-east-1.console.aws.amazon.com/secretsmanager/listsecrets?region=us-east-1

Secrets

Filter secrets by name, description, tag key, tag value, owning service or primary Region

Secret name	Description	Last retrieved (UTC)
/cafe/dbPassword	-	-
/cafe/dbUser	-	-
/cafe/dbName	-	-
/cafe/dbUrl	-	-
/cafe/currency	-	-
/cafe/timeZone	-	-
/cafe/showServerInfo	-	-

CloudShell Feedback

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2:12 PM 25-Oct-2025

25-26.

us-east-1.console.aws.amazon.com/secretsmanager/secret?name=%2Fcafe%2FdbPassword®ion=us-east-1

Encryption key
aws/secretsmanager

Secret name
/cafe/dbPassword

Secret ARN
arn:aws:secretsmanager:us-east-1:315224400642:secret:/cafe/dbPassword-2uowGI

Secret description
-

Overview Rotation Versions Replication Tags

Secret value Info

Retrieve and view the secret value.

Key/value Plaintext

Lab123#

Resource permissions - optional Info

Edit permissions

CloudShell Feedback

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27.

The screenshot shows a web browser at the top with several tabs, including 'Lab 038 - EC2: Create...', 'Course Invitation...', 'Challenge (Cafe) lab...', and 'CafeWebServer - All...'. The browser address bar shows 'us-east-1.console.aws.amazon.com/cloud9/ide/90f4f98cf6b14c1c80254eff92c4accf?region=us-east-1#'. Below the browser is a code editor with a dark theme. The editor has a sidebar on the left with a file explorer showing a project structure with folders like 'html', 'cafe', 'Aws', 'css', 'GuzzleHttp', 'Images', 'JmesPath', 'Par', and 'Http'. The main editor area shows a PHP file named 'getAppParameters.php' with the following code:

```
1 <?php
2
3 require "aws.phar"; $AWSSDK
4 require "aws-autoloader.php";
5 $ch = curl_init();
6
7 // get a valid TOKEN
8 $headers = array_f
```

Below the code editor is a terminal window with the following output:

```
bash - lp-10-0-0-215.ec2: x Immediate x
{
  "VersionId": "16646cbe-5dee-4059-b95b-790be402e104"
}
{
  "ARN": "arn:aws:secretsmanager:us-east-1:315224400642:secret:/cafe/dbUser-FFXPIm",
  "Name": "/cafe/dbUser",
  "VersionId": "6f1e6165-c1e2-4e03-b6ba-e7286c5b2aa5"
}
{
  "ARN": "arn:aws:secretsmanager:us-east-1:315224400642:secret:/cafe/dbPassword-2uowGI",
  "Name": "/cafe/dbPassword",
  "VersionId": "8458b7e9-8b15-479b-b26b-9f8b33746434"
}
}
Application Secrets Setup script completed.
voclabs:/environment/setup $ cd ../db/
voclabs:/environment/db $ ./set-root-password.sh
ERROR 2002 (HY000): Can't connect to local server through socket '/var/lib/mysql/mysql.sock' (2)
Set admin Password script completed.
Please check the set-root-password.log file to verify successful execution.
voclabs:/environment/db $ ./create-db.sh
ERROR 2002 (HY000): Can't connect to local server through socket '/var/lib/mysql/mysql.sock' (2)
Create Database script completed.
Please check the create-db.log file to verify successful execution.
voclabs:/environment/db $
```

The bottom of the image shows a Windows taskbar with a search bar, several application icons, and a system tray showing '29°C Sunny' and '2:15 PM 25-Oct-2025'.

29.

Lab 038 - EC2: Crea... Course Invitation - b... Challenge (Cafe) lab... c182747a470742211... /cafe/dbPassword... 44.200.220.107... CafeWebServer - All... us-east-1.console.amazonaws.com/cloud9/ide/90f4f98cf6b14c1c80254ef92c4accf?region=us-east-1#

File Edit Find View Go Run Tools Window Support Preview Run

Go to Anything (Ctrl-P)

html

- cafe
 - Aws
 - css
 - GuzzleHttp
 - Images
 - ImagesPath
 - Par
 - Http
 - Client
 - Message
 - aws-autoloader
 - aws.phar
 - aws.zip
 - CHANGELOG.n
 - getAppParam
 - index.php
 - LICENSE
 - menu.php
 - NOTICE
 - orderHistory.p
 - processOrder.p
 - README.md
 - serverInfo.php
 - index.html
 - cafe.zip
 - db.zip
 - README.md

AWS profile: default

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Welcome x index.html x index.php x getParameters.ph x

```
1 <?php
2
3 require 'aws.phar'; $awsSDK
4 require 'aws-autoloader.php';
5 $ch = curl_init();
6
7 // get a valid TOKEN
8 $headers = array()
```

(28 Bytes) 4:1 PHP Spaces: 2

mysql -i'p-10-0-0-215.ec2 x Immediate

```
Oct 25 09:19:30 ip-10-0-0-215.ec2.internal mariadb-prepare-db-dir[745]
Oct 25 09:19:30 ip-10-0-0-215.ec2.internal mariadb-prepare-db-dir[745]
Oct 25 09:19:30 ip-10-0-0-215.ec2.internal mariadb-prepare-db-dir[745]
Oct 25 09:19:30 ip-10-0-0-215.ec2.internal mariadb-prepare-db-dir[745]
Oct 25 09:19:30 ip-10-0-0-215.ec2.internal mariadb-prepare-db-dir[745]
Oct 25 09:19:31 ip-10-0-0-215.ec2.internal systemd[1]: Started mariad
voelabs:~/environment/db $ cd ~/environment/db
voelabs:~/environment/db $ ./set-root-password.sh

Set admin Password script completed.
Please check the set-root-password.log file to verify successful execution.

voelabs:~/environment/db $ ./create-db.sh

Create Database script completed.
Please check the create-db.log file to verify successful execution.

voelabs:~/environment/db $ mysql -u admin -p
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 5
Server version: 10.11.13-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

30.

Lab 038 - EC2: Crea... Course Invitation - b... Challenge (Cafe) lab... c182747a470742211... /cafe/dbPassword... 44.200.220.107... CafeWebServer - All... us-east-1.console.amazonaws.com/cloud9/ide/90f4f98cf6b14c1c80254ef92c4accf?region=us-east-1#

File Edit Find View Go Run Tools Window Support Preview Run

Go to Anything (Ctrl-P)

html

- cafe
 - Aws
 - css
 - GuzzleHttp
 - Images
 - ImagesPath
 - Par
 - Http
 - Client
 - Message
 - aws-autoloader
 - aws.phar
 - aws.zip
 - CHANGELOG.n
 - getAppParam
 - index.php
 - LICENSE
 - menu.php
 - NOTICE
 - orderHistory.p
 - processOrder.p
 - README.md
 - serverInfo.php
 - index.html
 - cafe.zip
 - db.zip
 - README.md

AWS profile: default

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Welcome x index.html x index.php x getParameters.ph x

```
1 <?php
2
3 require 'aws.phar'; $awsSDK
4 require 'aws-autoloader.php';
5 $ch = curl_init();
6
7 // get a valid TOKEN
8 $headers = array()
```

(28 Bytes) 4:1 PHP Spaces: 2

mysql -i'p-10-0-0-215.ec2 x Immediate

```
Database changed
MariaDB [cafe_db]> show tables;
+-----+
| Tables_in_cafe_db |
+-----+
| order              |
| order_item         |
| product             |
| product_group      |
+-----+
4 rows in set (0.000 sec)

MariaDB [cafe_db]> select * from product;
+----+ product_name | description | price | product_group | image_url |
+----+-----+-----+-----+-----+-----+
| 1 | Croissant | Fresh, buttery and fluff... Simply delicious! | 1.50 | 1 | Images/Croissants.jpg |
| 2 | Donut | We have more than half-a-dozen flavors! | 1.00 | 1 | Images/Donuts.jpg |
| 3 | Chocolate Chip Cookie | Made with Swiss chocolate with a touch of Madagascar vanilla | 2.50 | 1 | Images/Chocolate-Chip-Cookies.jpg |
| 4 | Muffin | Banana bread, blueberry, cranberry or apple | 3.00 | 1 | Images/Muffins.jpg |
| 5 | Strawberry Blueberry Tart | Bursting with the taste and aroma of fresh fruit | 3.50 | 1 | Images/Strawberry-Blueberry-Tarts.jpg |
| 6 | Strawberry Tart | Made with fresh ripe strawberries and a delicious whipped cream | 3.50 | 1 | Images/Strawberry-Tarts.jpg |
| 7 | Coffee | Freshly ground black or blended Colombian coffee | 3.00 | 2 | Images/Coffee.jpg |
| 8 | Hot Chocolate | Rich and creamy, and made with real chocolate | 3.00 | 2 | Images/Cup-of-Hot-Chocolate.jpg |
| 9 | Latte | Offered hot or cold and in various delicious flavors | 3.50 | 2 | Images/Latte.jpg |
+----+-----+-----+-----+-----+-----+
9 rows in set (0.000 sec)

MariaDB [cafe_db]> exit;
```

31.

The screenshot shows the AWS Cloud9 IDE interface. The left sidebar displays a file explorer with a project structure including folders for 'html', 'css', 'images', 'js', and 'php'. The main editor area shows a file named 'getAppParameters.php' with the following PHP code:

```
<?php
require 'aws.php'; #AWS SDK
require 'aws-autoloader.php';
$ch = curl_init();
// get a valid TOKEN
$headers = array (
```

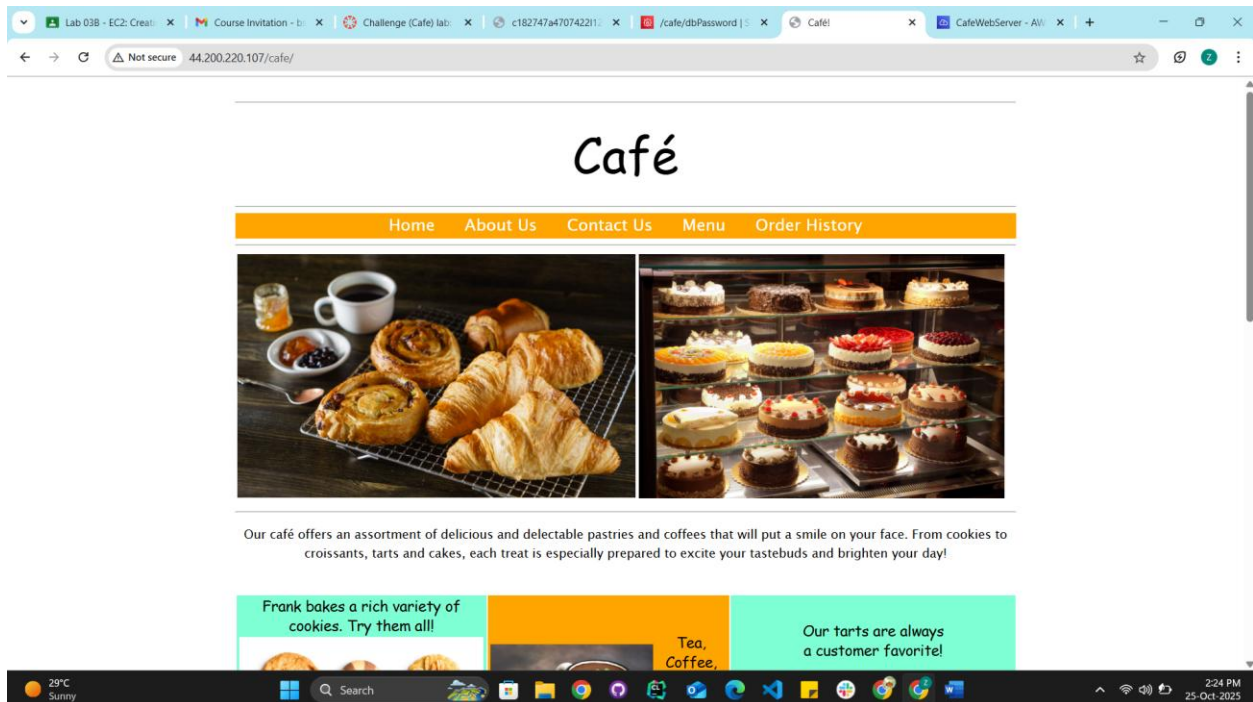
Below the editor, a terminal window is open with the following commands and output:

```
bash -lp-10-0-0-215.ec2: x Immediate
Tables_in_cafe_db
+-----+
| order |
| order_item |
| product |
| product_group |
+-----+
4 rows in set (0.000 sec)

MariaDB [cafe_db]> select * from product;
+----+-----+-----+-----+-----+-----+
| id | product_name | description | price | product_group | image_url |
+----+-----+-----+-----+-----+-----+
| 1 | Croissant | Fresh, buttery and fluffy... Simply delicious! | 1.50 | 1 | images/Croissants.jpg |
| 2 | Donut | We have more than half-a-dozen flavors! | 1.00 | 1 | images/Donuts.jpg |
| 3 | Chocolate Chip Cookie | Made with Swiss chocolate with a touch of Madagascar vanilla | 2.50 | 1 | images/Chocolate-Chip-Cookies.jpg |
| 4 | Muffin | Banana bread, blueberry, cranberry or apple | 3.00 | 1 | images/Muffins.jpg |
| 5 | Strawberry Blueberry Tart | Bursting with the taste and aroma of fresh fruit | 3.50 | 1 | images/Strawberry-Blueberry-Tarts.jpg |
| 6 | Strawberry Tart | Made with fresh ripe strawberries and a delicious whipped cream | 3.50 | 1 | images/Strawberry-Tarts.jpg |
| 7 | Coffee | Freshly-ground black or blended Colombian coffee | 3.00 | 2 | images/Coffee.jpg |
| 8 | Hot Chocolate | Rich and creamy, and made with real chocolate | 3.00 | 2 | images/Cup-of-Hot-Chocolate.jpg |
| 9 | Latte | Offered hot or cold and in various delicious flavors | 3.50 | 2 | images/Latte.jpg |
+----+-----+-----+-----+-----+-----+
9 rows in set (0.000 sec)

MariaDB [cafe_db]> exit;sudo sed -i "21 date.timezone = \"America/New_York\" " /etc/php.ini
Bye
voclabs:~/environment/db $ sudo sed -i "21 date.timezone = \"America/New_York\" " /etc/php.ini
sudo service httpd restartvoclabs:~/environment/db $
```

32.



us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instances:

EC2 > Instances

Successfully attached CafeRole to Instance i-043b2a94c2f3fbdef

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
aws-cloud9-Ca...	i-043b2a94c2f3fbdef	Running	t2.micro	2/2 checks passed		us-east-1a	ec2-44-200

i-043b2a94c2f3fbdef (aws-cloud9-CafeWebServer-90f4f98cf6b14c1c80254eff92c4accf)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-043b2a94c2f3fbdef	Public IPv4 address 44.200.220.107 open address	Private IPv4 addresses 10.0.0.215
IPv6 address -	Instance state Running	Public DNS ec2-44-200-220-107.compute-1.amazonaws.com open address

CloudShell Feedback

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2:31 PM 25 Oct 2025

44.200.220.107/cafe/menu.php

Café

Home Menu Order History

Pastries



Croissant
\$1.50
Fresh, buttery and fluffy... Simply delicious!
Quantity:



Donut
\$1.00
We have more than half-a-dozen flavors!
Quantity:



Chocolate Chip Cookie
\$2.50
Made with Swiss chocolate with a touch of Madagascar vanilla
Quantity:







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2:32 PM 25 Oct 2025

Task 5: Testing the web application

34-37.

Cafe

Home Menu **Order History**

Order History

Order Number: 4 Date: 2025-10-25 Time: 09:35:09 Total Amount: \$17.50

Item	Price	Quantity	Amount
Chocolate Chip Cookie	\$2.50	7	\$17.50

Order Number: 3 Date: 2025-10-25 Time: 09:33:53 Total Amount: \$10.00

Item	Price	Quantity	Amount
Chocolate Chip Cookie	\$2.50	4	\$10.00

Order Number: 2 Date: 2025-10-25 Time: 09:33:07 Total Amount: \$9.00

Item	Price	Quantity	Amount
Croissant	\$1.50	6	\$9.00

Order Number: 1 Date: 2025-10-25 Time: 09:32:58 Total Amount: \$3.00

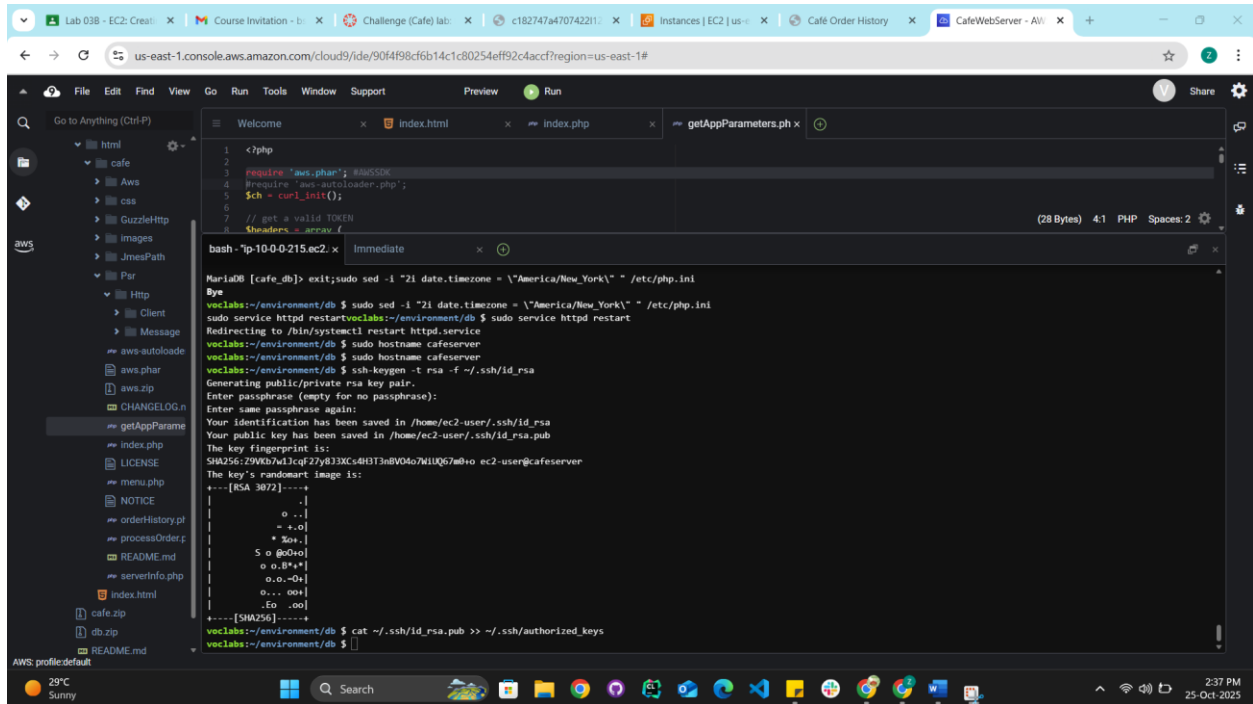
Item	Price	Quantity	Amount
Donut	\$1.00	3	\$3.00

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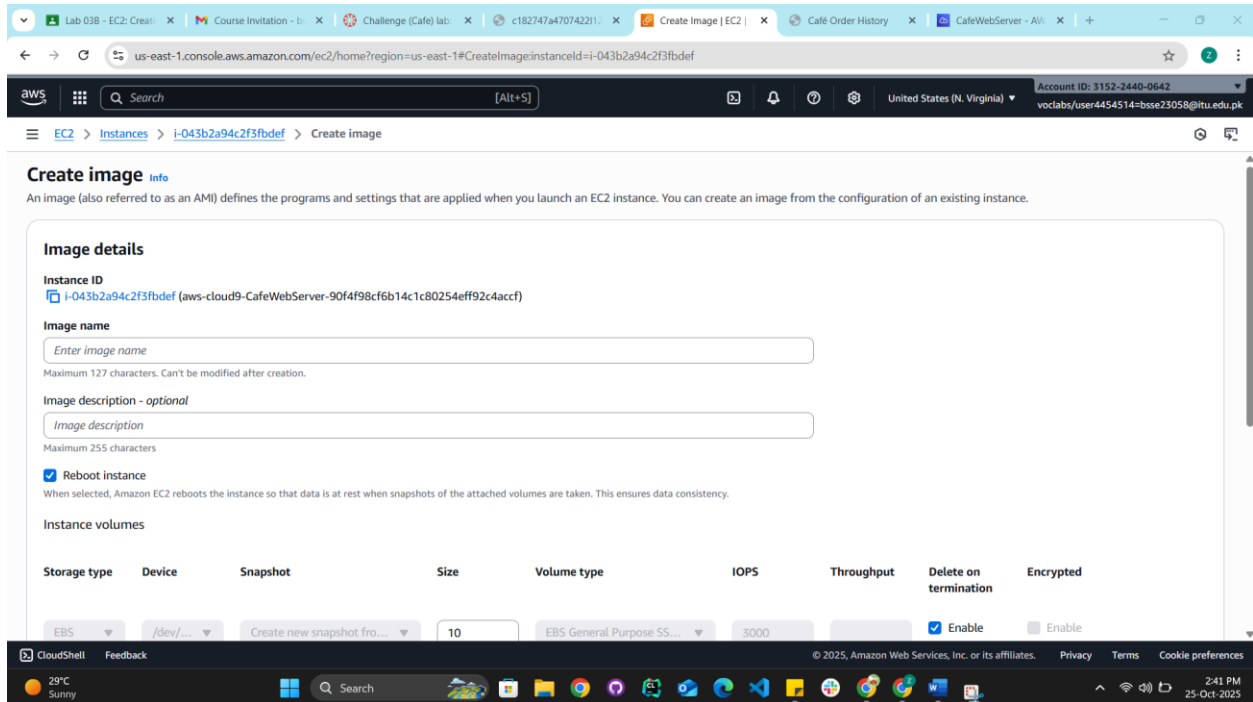
New business requirement: Creating development and production websites in different AWS Regions (challenge #3)

Task 6: Creating an AMI and launching another EC2 instance

38-40.



41-42.



43, 44.

Submit

Question 5: When you create an AMI from an instance, will the instance be rebooted?

- ☐ Yes, always
- ☐ No, never
- ☒ You have the option *not* to reboot, but by default it will be rebooted
- ☐ You have the option to reboot, but by default it will *not* be rebooted

Submit

Question 6: In what ways can you modify the root volume properties when you create an AMI from an instance?

- ☐ You cannot change the root volume details.
- ☐ You can edit the size, but nothing else.
- ☒ You can edit the size and 'delete on termination' setting, but not the volume type.
- ☐ You can edit the size and volume type, but not the 'delete on termination' setting.

Submit

Question 7: Can you add more volumes to an AMI that you create from an instance that only has one volume?

- ☒ Yes
- ☐ No

Submit

45-46.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

Account ID: 3152-2440-0642
voclabs/user4454514-bbse23058@itu.edu.pk

EC2 > Instances

Dashboard
EC2 Global View
Events

▼ Instances

Instances

Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager [New](#)

▼ Images

AMIs
AMI Catalog

▼ Elastic Block Store

Volumes

Currently creating AMI ami-05cd3f200d81b6c8c from instance i-043b2a94c2f3fbdef. Check that the AMI status is 'Available' before deleting the instance or carrying out other actions related to this AMI.

Instances (1/1) [Info](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☒	aws-cloud9-Ca...	i-043b2a94c2f3fbdef	Running	t2.micro	2/2 checks pass	View alarms	us-east-1a	ec2-44-200

i-043b2a94c2f3fbdef (aws-cloud9-CafeWebServer-90f4f98cf6b14c1c80254eff92c4accf)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance summary [Info](#)

Instance ID i-043b2a94c2f3fbdef	Public IPv4 address 44.200.220.107 open address	Private IPv4 addresses 10.0.0.215
IPv6 address -	Instance state Running	Public DNS ec2-44-200-220-107.compute-1.amazonaws.com open address

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47.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#images:visibility=owned-by-me

Account ID: 3152-2440-0642
voclabs/user4454514-bsse23058@itu.edu.pk

EC2 > AMIs

Amazon Machine Images (AMIs) (1)

Owned by me | Find AMI by attribute or tag

AMI ID	Source	Owner	Visibility	Status	Creation time
ami-05cd3f200d81b6c8c	315224400642/CafeServer	315224400642	Private	Available	2025/10/25 2:47 PM

Select an AMI

48.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#images:visibility=owned-by-me

Account ID: 3152-2440-0642
voclabs/user4454514-bsse23058@itu.edu.pk

EC2 > AMIs

Amazon Machine Images (AMIs) (1/1)

Owned by me | Find AMI by attribute or tag

Name	AMI name	AMI ID	Source	Owner	Visibility
☒	CafeServer	ami-05cd3f200d81b6c8c	315224400642/CafeServer	315224400642	Private

AMI ID: ami-05cd3f200d81b6c8c

Details | Permissions | Storage | My AMI usage | Tags

Attribute	Value
AMI ID	ami-05cd3f200d81b6c8c
AMI name	CafeServer
Root device name	/dev/xvda
Image type	machine
Owner account ID	315224400642
Status	Available
Platform details	Linux/UNIX
Architecture	x86_64
Source	315224400642/CafeServer
Root device type	EBS
Usage operation	RunInstances
Virtualization type	hvm

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#images:visibility=owned-by-me:imageid=ami-0a52211cf6e335f3c

aws [Search] [Alt+S] United States (Oregon) Account ID: 3152-2440-0642 voclabs/user4454514-bsse23058@itu.edu.pk

EC2 > AMIs

Amazon Machine Images (AMIs) (1) Info

Owned by me Find AMI by attribute or tag

AMI ID = ami-0a52211cf6e335f3c Clear filters

AMI name	AMI ID	Source	Owner	Visibility	Status
CafeServer	ami-0a52211cf6e335f3c	315224400642/CafeServer	315224400642	Private	Available

Select an AMI

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49-50.

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#LaunchInstances:

aws [Search] [Alt+S] United States (Oregon) Account ID: 3152-2440-0642 voclabs/user4454514-bsse23058@itu.edu.pk

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0d7d85a6102b24de8)

Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing and free tier usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
[Create EBS snapshot policy](#)

Manage detailed monitoring

Create Load Balancer

Create AWS budget

Manage CloudWatch alarms

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51.

Instance summary for i-0d7d85a6102b24de8 (ProdCafeServer) info

Updated less than a minute ago

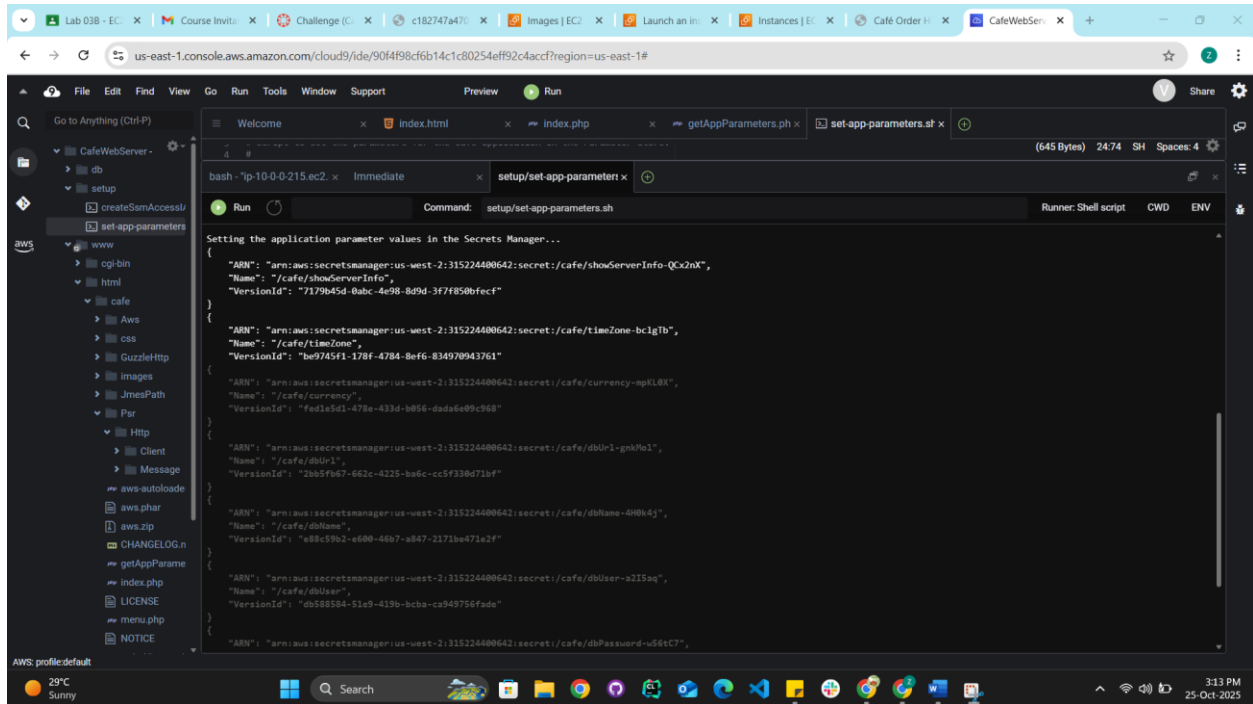
Instance ID i-0d7d85a6102b24de8	Public IPv4 address 54.191.230.149 open address	Private IPv4 addresses 10.0.0.161
IPv6 address -	Instance state Running	Public DNS ec2-54-191-230-149.us-west-2.compute.amazonaws.com open address
Hostname type IP name: ip-10-0-0-161.us-west-2.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-0-161.us-west-2.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.small	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendation Learn more
Auto-assigned IP address 54.191.230.149 [Public IP]	VPC ID vpc-05a93edbf35c6ac3a (Lab VPC Region 2)	Auto Scaling Group name -
IAM Role CafeRole	Subnet ID subnet-008345b1b6ccf3ff (Public Subnet)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-west-2:315224400642:instance/i-0d7d8	

52-55.

```

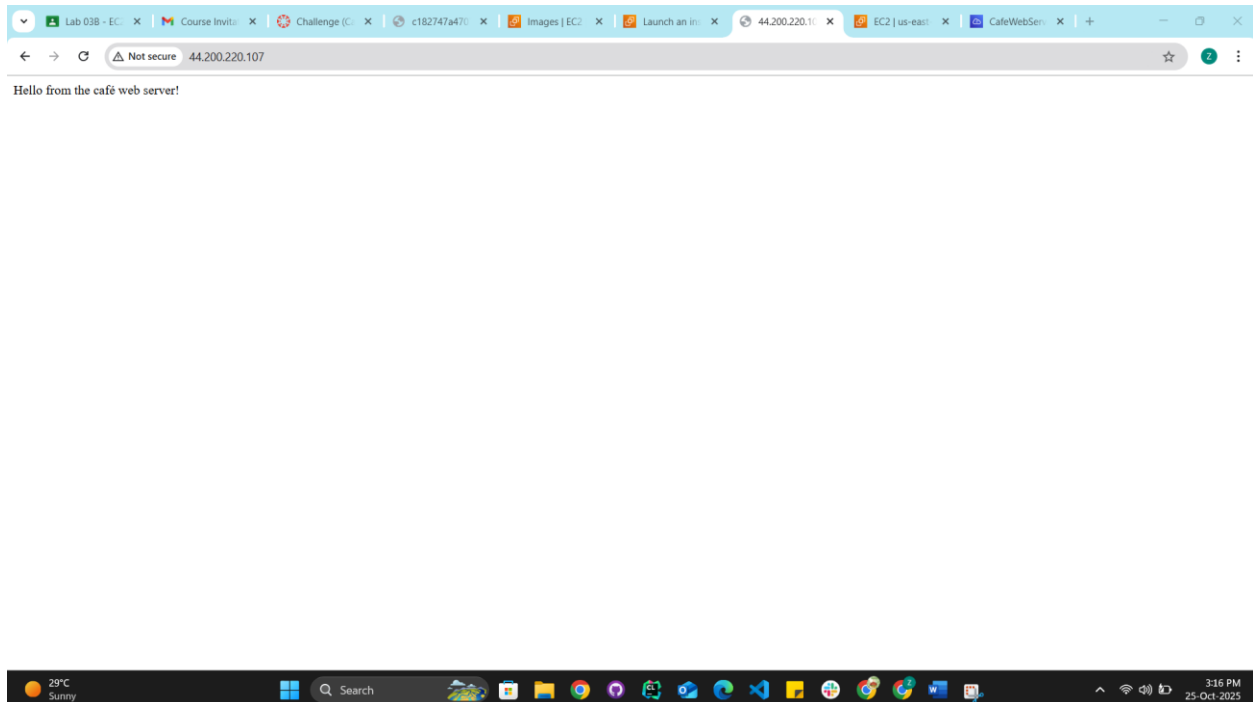
1 # Script to set up application parameters and secrets for the CafeWebServer.
2 #
3 # Get the region where the instance is running, and set as it the default AWS region.
4 # This ensures that we are using the Parameter Store in the region where the instance is running.
5
6 echo "Setting the default AWS region..."
7 TOKEN=$(curl -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")
8 #curl -H "X-aws-ec2-metadata-token: $TOKEN" -v http://169.254.169.254/latest/meta-data/
9
10 echo "${curl -H "X-aws-ec2-metadata-token: $TOKEN" -v http://169.254.169.254/latest/meta-data/placement/availability zone}"
11
12 region="us-west-2"
13 export AWS_DEFAULT_REGION=$region
14 echo "Region -> $AWS_DEFAULT_REGION"
15
16 # Set the application parameter values.
17 #
18 publicDNS="ec2-44-208-220-187.compute-1.amazonaws.com"
19 echo "Public DNS -> $publicDNS"
20
21 echo "Setting the application parameter values in the Secrets Manager..."
22 aws secretsmanager create-secret --name "/cafe/timeZone" --secret-string "America/New York"
23 aws secretsmanager create-secret --name "/cafe/currency" --secret-string "$"
24 aws secretsmanager create-secret --name "/cafe/dbUser" --secret-string "publicDNS"
25 aws secretsmanager create-secret --name "/cafe/dbName" --secret-string "cafe_db"
26 aws secretsmanager create-secret --name "/cafe/dbUser" --secret-string "admin"
27 aws secretsmanager create-secret --name "/cafe/dbPassword" --secret-string "Lab123M"
28
29 echo
30 echo "Application Secrets Setup script completed."
31
32
33 bash -lp 10-0-0-215.ec2.x Immediate
34
35 voclabs:/environment $
  
```

56.



Task 7: Verifying the new café instance

57.



Café

[Home](#) [About Us](#) [Contact Us](#) [Menu](#) [Order History](#)



Our café offers an assortment of delicious and delectable pastries and coffees that will put a smile on your face. From cookies to croissants, tarts and cakes, each treat is especially prepared to excite your tastebuds and brighten your day!

Frank bakes a rich variety of cookies. Try them all!

Tea, Coffee,

Our tarts are always a customer favorite!

Café

[Home](#) [Menu](#) [Order History](#)

Pastries



Croissant

\$1.50

Fresh, buttery and fluffy... Simply delicious!

Quantity:



Donut

\$1.00

We have more than half-a-dozen flavors!

Quantity:



Chocolate Chip Cookie

\$2.50

Made with Swiss chocolate with a touch of Madagascar vanilla

Quantity:



Lab 03B - EC2: Crea...Course Invitation - b...Challenge (Cafe) lab...c182747a470742211...Instances | EC2 | us-...Cafe Order HistoryCafeWebServer - All...

← → ↻ Not secure 44.200.220.107/cafe/orderHistory.php ☆ ⓘ 🔒 ⋮

Cafe

HomeMenuOrder History

Order History

Order Number: 4Date: 2025-10-25Time: 09:35:09Total Amount: \$17.50

Item	Price	Quantity	Amount
Chocolate Chip Cookie	\$2.50	7	\$17.50

Order Number: 3Date: 2025-10-25Time: 09:33:53Total Amount: \$10.00

Item	Price	Quantity	Amount
Chocolate Chip Cookie	\$2.50	4	\$10.00

Order Number: 2Date: 2025-10-25Time: 09:33:07Total Amount: \$9.00

Item	Price	Quantity	Amount
Croissant	\$1.50	6	\$9.00

Order Number: 1Date: 2025-10-25Time: 09:32:58Total Amount: \$3.00

Item	Price	Quantity	Amount
Donut	\$1.00	3	\$3.00

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29°C
Sunny





2:35 PM
25-Oct-2025

GRADES:

Challenge (Cafe) lab: Creating a Dynamic Website for the Cafe

Due No Due Date Points 30 Submitting an external tool

00:58 ▶ Start Lab ■ End Lab AWS Details Details

Submit Submission Report Grades

EN_US

- Tested the web application
- Created an AMI
- Deployed a second copy of the web application to another AWS Region

Submitting your work

63. At the top of these instructions, choose **Submit** to record your progress, and when prompted, choose **Yes**.

64. If the results don't display after a couple of minutes, return to the top of these instructions and choose **Grades**.

Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is what will be recorded for this lab.

Total score	23/30
[Answer 01]	1/1
[Answer 02]	1/1
[Answer 03]	1/1
[Answer 04]	0/1
[Task 3] Web server SG config	5/5
[Task 4] Instance permissions	5/5

◀ Previous Next ▶

29°C Sunny 3:31 PM 25-Oct-2025

Challenge (Cafe) lab: Creating a Dynamic Website for the Cafe

Due No Due Date Points 30 Submitting an external tool

00:58 ▶ Start Lab ■ End Lab AWS Details Details

Submit Submission Report Grades

EN_US

- Tested the web application
- Created an AMI
- Deployed a second copy of the web application to another AWS Region

Submitting your work

63. At the top of these instructions, choose **Submit** to record your progress, and when prompted, choose **Yes**.

64. If the results don't display after a couple of minutes, return to the top of these instructions and choose **Grades**.

Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is what will be recorded for this lab.

[Answer 06]	1/1
[Answer 07]	1/1
[Task 6B] Instance role	3/3
[Task 6C] Instance type	2/2
[Task 6D] From custom AMI	0/3
[Task 7] Menu page loads	0/3

◀ Previous Next ▶

29°C Sunny 3:32 PM 25-Oct-2025

SUBMISSION REPORT:

The screenshot displays the AWS Academy interface for a challenge lab. The browser address bar shows the URL: `awsacademy.instructure.com/courses/140296/assignments/1630583`. The page title is "Challenge (Cafe) lab: Creating a Dynamic Website for the Café". The submission report is titled "Submission Report" and shows the execution time as "Sat Oct 25 3:31:04 PDT 2025". The report contains a list of tasks and their results:

- [Answer 01] Correct, the instance is running in a public subnet.
- [Answer 02] Correct, the instance does have a public IP address assigned.
- [Answer 03] Correct, only TCP port 22 is open, and only to a specific range of IP addresses.
- [Answer 04] That is incorrect, the EC2 instance does not have an IAM role attached.
- [Task 3] SUCCESS! The web server instance's security group has port 80 open.
- [Task 4] SUCCESS! The CafeRole has been attached to the EC2 instance.
- [Answer 05] Correct, you have the option to not reboot when creating an AMI, but by default it will be rebooted.
- [Answer 06] Correct, You can edit the size and 'delete on termination' setting, but not the volume type.
- [Answer 07] Correct, you can add more volumes to an AMI that you create from an instance with only one volume.

Task 64: If the results don't display after a couple of minutes, return to the top of these instructions and choose **Grades**.

Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is what will be recorded for this lab.

The submission report also includes a section for "Task 7" with the following details:

```
[default]
region = us-east-1
gradeFile = /mnt/vocwork4/grader/eee_g_2990563/asn4707421_3/asn4707422_1/tmp/temp_uf_10252025/.VK9Uck
reportFile = /mnt/vocwork4/grader/eee_g_2990563/asn4707421_3/asn4707422_1/tmp/temp_uf_10252025/.yZMQ27
```

The submission report also includes a section for "Task 7" with the following details:

```
[Task 7] Menu page loads
```

Lab 03B - EC2 x Course Invit... x Challenge (C... x 6aar7zwfr0... x Instances | EC... x Café Menu x Instances | EC... x CafeWebServ... x Café Order H... x

awsacademy.instructure.com/courses/140296/assignments/1630583

ACAv3EN-US-LT113-140296 > Assignments > Challenge (Cafe) lab: Creating a Dynamic Website for the Café
> Challenge (Cafe) lab: Creating a Dynamic Website for the Café

Home
Discussions
Grades
Lucid (Whiteboard)
Assignments

Challenge (Cafe) lab: Creating a Dynamic Website for the Café

Due No Due Date Points 30 Submitting an external tool

Submission Report

```
webServerInstance=i-043b2a94c2f3fbdef
webServerInstanceSg = sg-0ace58a4964200611
sgOpenPorts = 80
1st port found open (should be 22): 80
2nd port found open (should be 80): 22
3rd port found open (should be none):
iPro=arn:aws:iam::315224400642:instance-profile/CafeRole
cRole=CafeRole
```

64. If the results don't display after a couple of minutes, return to the top of these instructions and choose **Grades**.

Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is what will be recorded for this lab.

[Task 7] Menu page loads

29°C Sunny

Lab 03B - EC2 x Course Invit... x Challenge (C... x 6aar7zwfr0... x Instances | EC... x Café Menu x Instances | EC... x CafeWebServ... x Café Order H... x

awsacademy.instructure.com/courses/140296/assignments/1630583

Challenge (Cafe) lab: Creating a Dynamic Website for the Café

Home
Discussions
Grades
Lucid (Whiteboard)
Assignments

Challenge (Cafe) lab: Creating a Dynamic Website for the Café

Due No Due Date Points 30 Submitting an external tool

Submission Report

```
acctId=315224400642
newRegionOneAmi=ami-05cd3f200d81b6c8c
r2subnetId=subnet-008345b1b6ccf33ff
r2subnetName=Public

webServerNewRegionInstance=i-0d7d85a6102b24de8
instPro=CafeRole
r2instanceType=t2.small

newRegionTwoAmi=ami-0a52211cf6e335f3c
```

64. If the results don't display after a couple of minutes, return to the top of these instructions and choose **Grades**.

Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is what will be recorded for this lab.

[Task 7] Menu page loads

29°C Sunny